Krithik Krishna K U *Digital Design Engineer*

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Profile

Digital Design Engineer specializing in **High-Speed FPGA architectures** and **RF systems** for mission-critical Defense and Space projects. Expert in custom IP development for LO synthesis and data conversion, with a unique capability in Full-Stack GUI implementation (Front-end/Back-end) for hardware control. Proven track record in Python-based automation and hardware validation, delivering end-to-end solutions from high-speed logic to user interfaces.

Professional Experience

DIGITAL Engineer
06/2024 – present | Bengaluru, India

RFMW Innovations Lab

- **High-Speed RF IP Development:** Architect and deploy digital IPs for high-performance RF components, optimizing high-speed data conversion and frequency synthesis.
- **Ai-Accelerated Tooling:** Developing Node.js and Python scripts integrated with AI acceleration to automate complex hardware orchestration and real-time data analysis, enhancing system intelligence.
- **Design Automation & Efficiency:** Engineered Python-based frameworks to auto-generate repetitive Verilog RTL, minimizing manual coding errors and drastically accelerating the FPGA implementation cycle.
- **Full-Stack GUI Development:** Built comprehensive front-end and back-end custom GUIs (Python/Tkinter) for real-time register control and hardware telemetry, streamlining the validation of mission-critical systems.
- **System-Level RF Design:** Lead the development of Up/Down conversion modules, integrating PLL-based LO generation, multi-band SPXT RF switching, digital attenuators, and Power Amplifiers.
- **Cross-Platform FPGA Engineering:** Hands-on experience developing for Xilinx, Intel Altera, and Microsemi FPGA families, ensuring hardware-agnostic performance for defense and space applications.
- **Hardware Validation & Debugging:** Expert use of lab instrumentation (VNAs, Spectrum Analyzers, Signal Generators) for hands-on debugging of high-frequency RF systems.
- **Strategic Defense Contributions:** Deliver high-reliability solutions in collaboration with ISRO, DRDO, BEL, and CABS for national-level defense projects.

Education

Amrita Viswa Vidyapeetham
07/2023 – 07/2025 | Coimbatore, India

M.Tech in VLSI Design

Completed a postgraduate program with a strong focus on digital and analog VLSI systems, RTL design, HDL coding and ASIC/FPGA implementation. Gained hands-on experience with industry-standard EDA tools like Xilinx Vivado and Synopsys Prime Time. Worked on lab-based projects involving RTL design, logic synthesis, and physical design flow.

Christ College of Engineering
08/2018 – 06/2022 | Thrissur, India

B.TECH in Electrical and Electronics

Enhanced troubleshooting and debugging skills in electronic circuits while collaborating on interdisciplinary projects that integrated both hardware and software components, strengthening teamwork and project management abilities.

Core Technical Skills

FPGA Design & Architecture
SystemVerilog, RTL Design, Timing Analysis, IP Development.

High-Speed Signal Chains
High-Speed DAC/PLL Configuration, Frequency Synthesis.

Peripheral & Power Control
Integration of Sensors in mixed-signal peripherals.

Automated RTL Generation
Python-based Meta-Programming for Verilog/SystemVerilog.

Hardware Interface & Control
Full-Stack GUI Development, Real-Time Telemetry, and Protocol Implementation.

RF Domain Expertise
Design and integration of SPXT Switches, Digital Attenuators, RF Filters, and Power Amplifiers.

Platforms & Tools

FPGA Tools
Xilinx Vivado, Intel Quartus Prime, Microsemi Libero SoC.

Software & Version Control
Python Scripting, Node.js, HTML, GIT Version Control

Lab Instrumentation
VNA, Spectrum Analyzers, Signal Generators, Oscilloscopes.

Projects

End-to-End RF Up/Down Conversion System
06/2025 – Present

Full-Lifecycle RF Up/Down Conversion System Implementation Led the end-to-end development of high-performance conversion modules, integrating high-speed DACs and PLLs for LO generation and precision frequency synthesis. Designed FPGA-based control logic for SPXT switches, PAs, digital attenuators, and thermal sensors, ensuring compliance with mission-critical ISRO/DRDO standards. Streamlined system orchestration via a full-stack Node.js/HTML interface and automated RTL generation scripts, achieving a 98% reduction in manual coding errors while bridging the gap between digital architectures and RF front-ends.

Data-Driven RTL Automation & Framework
03/2025 – Present

Engineered a Python-based automation framework that parses Excel-based register maps to auto-generate synthesizable Verilog code for UART controllers. By bridging the gap between system specifications and hardware description, the tool reduced manual coding errors by 98% and significantly accelerated the RTL development cycle. Integrated a custom Node.js/HTML GUI to allow real-time modifications to packet structures, streamlining the validation process for complex digital-RF interfaces.

Design of CMOS based cascode mixer suitable for V-band application
06/2024 – present

Designed and optimized a CMOS-based cascode mixer for V-band applications, using a distributed topology with gate inductive positive feedback to enhance performance up to 60 GHz while addressing high-frequency operation and isolation challenges.

Phase synchronized frequency synthesizer using wide band PLL
06/2024 – 11/2024

Developed a GUI-controlled wide-band frequency synthesizer (44 MHz to 22.8 GHz) using PLLs and UART via USB Type-C. Programmed an Altera Cyclone V FPGA for frequency generation and phase synchronization across PLLs, utilizing LMK 04848 and LMX 2820 chips with SPI and UART protocols for communication.

Courses

RTL TO GDS
06/2024 – 11/2024

NPTEL

Gained proficiency in industry-standard EDA tools for RTL synthesis, place and route, and layout verification, while acquiring skills in digital circuit design, simulation, verification methodologies, and timing analysis to ensure robust circuit performance.

Industrial Automation and Industry 4.0
04/2020

Texas Instruments / BITS Pilani

Completed an intensive workshop on modern automation standards, focusing on smart manufacturing and integrated system architectures.

Languages

English — Fluent

Malayalam — Native/Bilingual

Hindi — Proficient

Tamil — Proficient